

CHANDAN SINGH

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PROFESSIONAL SUMMARY

B.Tech CSE student (Expected 2027) specializing in AI/ML engineering with hands-on experience building production-grade computer vision, NLP, and full-stack AI systems. Delivered deployable projects using YOLOv8, LSTM, XGBoost, and LLM APIs; competed in national and AMD-sponsored hackathons targeting \$20,000+ prize pools.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

AI/ML Frameworks: TensorFlow, PyTorch

Backend & Cloud: FastAPI, Flask, Docker, Git, HuggingFace Hub, ROCm

Full-Stack & DB: React, Vite, PostgreSQL, SQLAlchemy, JWT Authentication, Twilio API

Concepts: Computer Vision, NLP

TECHNICAL PROJECTS

NikaAI - AI Quality Copilot for MSME Manufacturing | github.com/chandanNegi3G671/NikaAI

- Developed an end-to-end AI defect inspection system for the AMD AI Hackathon, containerized with Docker Compose and deployed on AMD Developer Cloud with ROCm GPU acceleration.
- Fine-tuned YOLOv8s (83% mAP, 17 defect classes) and integrated with Gemma 4 LLM for automated root-cause analysis, severity assessment, and recommended corrective actions. Implemented MC Dropout for uncertainty quantification and a secure RBAC system (Admin/Supervisor/Worker) using JWT.

TRINETRA - AI Scam Intelligence Engine | github.com/chandanNegi3G671/TRINETRA

- Designed a real-time fraud detection platform achieving 95%+ accuracy using an ensemble model (XGBoost + Rule-based logic) to analyze URLs, SMS, and WhatsApp screenshots.
- Integrated Google Safe Browsing API, IP geolocation, and domain entropy scoring into a unified 0-100 threat score.
- Deployed a full-stack application with a Twilio WhatsApp bot for instant scam alerts; secured with rate limiting and input sanitization middleware.

Factory Defect Guard | huggingface.co/negi3G61/factory-defect-guard

- Fine-tuned YOLOv8s on 29,354 industrial images across 17 defect classes (Steel, PCBs, MVTec components) using a multi-stage training pipeline.
- Built a custom VOC-to-YOLO annotation converter; achieved 78.8% precision and 72.2% recall. Hosted model weights and inference interface on HuggingFace Hub for open-access industrial defect detection.

SIGNSPEAK - Real-Time Sign Language Recognition | github.com/chandanNegi3G671/SIGNSPEAK

- Built a word-level ASL recognition system using stacked LSTM networks (30-frame sequences) and MediaPipe landmark extraction.
- Developed a full-stack application with a React/Vite frontend and Flask API, supporting real-time caption display and vocabulary extension.
- Optimized inference with a 70% confidence threshold to ensure reliable real-time transcription.

ACHIEVEMENTS & HACKATHONS

AMD AI Hackathon Participant - Built NikaAI defect inspection system targeting \$20,000+ prize
Deployed production-grade AI systems with Docker, ROCm, GPU acceleration, and HuggingFace Hub
Built real-time fraud detection achieving 95%+ accuracy with ensemble ML models